

ASIC RTL Current-State Specification and Dataset Report

SPADMIC Digital RTL Database

Internal Engineering Specification Package

Evidence-based RTL, register-map, interface, verification, and readiness snapshot.

Repository	KarimSabra13/SPADMIC
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Active chip top	<code>spadmic_top_v1</code>
Axis TDC top	<code>mptdc_top_asic</code>
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1 Executive Summary

Current RTL position

The active SPADMIC digital chip integration is `spadmic_top_v1`. It integrates I2C control, global CSR sequencing, three MPTDC axes, shared TDC readout, position capture, correlated packet export, and an 8-bit source-synchronous DDR TX interface. The RTL is structured and documented, but final signoff is not yet complete because analog macro contracts, generated-clock constraints, CDC waivers, output timing, and coverage closure remain open.

Item	Value
Primary purpose	Chip-level digital integration for SPAD/TDC acquisition, position capture, control-plane access, and shared packet export.
Active top-level module	<code>\{\texttt{spadmic}\}_\{\texttt{top}\}_\{\texttt{v1}\}</code> ; confirmed by TOP filelist and active readiness graph.
Configuration method	I2C slave at 7-bit address 0x42, 12-bit CSR address, 32-bit CSR data.
Main functional blocks	I2C/CSR, global sequencer, TDC axes, position block, correlated TX, DDR TX.
RTL maturity	Functionally organized and documented; not final mixed-signal/tapeout signoff.
Verification maturity	Directed/VIP collateral exists; documented coverage closure is still incomplete.
Primary readiness blockers	PLL/SPAD/oscillator macro contracts, oscillator-domain STA, CDC waiver proof, DDR output timing, coverage closure.

Source: `TOP/filelist.f:37-39`; `TOP/docs/10_TAPEOUT_READINESS.md:14-24,26-49`; `TOP/rtl/spadmic_top_v1.sv:128-442`; `TOP/rtl/spadmic_pkg.sv:21-25`; `MPTDC/lab_snapshots/README.md:9-15,29-34`

2 Repository and Evidence Scope

The report scope is the active SPADMIC ASIC RTL database under TOP, MPTDC, and I2C. The OpenROAD checkout exists in the workspace but is not treated as active chip RTL because it is not referenced by the TOP, MPTDC, or I2C compile filelists inspected for this report.

Area	Primary paths	Role	Confidence
TOP	TOP/rtl, TOP/tb, TOP/docs, TOP/ci, TOP/scripts	Chip-level RTL, top verification collateral, active architecture/readiness documentation	High
MPTDC	MPTDC/rtl, MPTDC/tb, MPTDC/docs, MPTDC/results, MPTDC/lab_{}snapshots, MPTDC/syn	Preserved TDC kernel, calibration data, synthesis snapshots, verification collateral	High
I2C	I2C/rtl, I2C/docs, I2C/filelist.f	Reusable I2C control-plane slave and CSR bridge	High
Generated/report collateral	MPTDC/lab_{}snapshots, MPTDC/results, TOP/build	Metrics and status only when explicitly sourced	Medium

3 Top-Level Architecture

The active chip-level datapath is rooted at `spadmic_{}top_{}v1`. Legacy TOP modules remain in-tree but are documented as outside the active chip datapath. The hierarchy below is the current architectural view used for this specification.

```
spadmic_top_v1
|- spadmic_i2c_slave -> spadmic_i2c_csr_bridge -> spadmic_csr_decoder
|- spadmic_global_csr -> spadmic_top_sequencer
|- spadmic_tdc_axis_wrapper x3 -> spadmic_ref_stop_qualifier + mptdc_top_asic
|   '- mptdc_core -> async frontend, oscillators, PD matrix, CDC, FIFO/readout
|- spadmic_tdc_shared_readout
|- spadmic_position_block -> spadmic_axis_cluster_scan x3 + frame FIFO
|- spadmic_correlated_tx -> 2048 x 16 output FIFO
'- spadmic_ddr_tx -> chip_tx_clk_o/chip_tx_valid_o/chip_tx_data_o[7:0]
```

Block	Role	Evidence
<code>spadmic_{}top_{}v1</code>	Chip-level integration shell	<code>TOP/rtl/spadmic_{}top_{}v1.sv:11-444</code>
<code>spadmic_{}i2c_{}slave</code>	Pointer-based I2C front end	<code>I2C/rtl/spadmic_{}i2c_{}slave.sv:11-380</code>
<code>spadmic_{}i2c_{}csr_{}bridge</code>	Single-outstanding I2C-to-CSR transaction bridge	<code>I2C/rtl/spadmic_{}i2c_{}csr_{}bridge.sv:11-89</code>
<code>spadmic_{}csr_{}decoder</code>	12-bit TOP CSR region decoder with read timeout	<code>TOP/rtl/spadmic_{}csr_{}decoder.sv:95-129,165-213</code>
<code>spadmic_{}global_{}csr</code>	Requested control image, status, and global faults	<code>TOP/rtl/spadmic_{}global_{}csr.sv:82-190</code>
<code>spadmic_{}top_{}sequencer</code>	Draining requested-to-active control sequencer	<code>TOP/rtl/spadmic_{}top_{}sequencer.sv:40-116</code>
<code>spadmic_{}tdc_{}axis_{}wrapper x3</code>	Per-axis SPAD/CAL gating, stop qualifier, and MPTDC wrapper	<code>TOP/rtl/spadmic_{}tdc_{}axis_{}wrapper.sv:43-87</code>

Block	Role	Evidence
mptdc_{top_{asic} x3}	Pad-facing MPTDC integration boundary	MPTDC/rtl/top/mptdc_{top_{asic}.sv:25-174
mptdc_{core} x3	Vernier measurement kernel, CDC, context bank, FIFO/readout	MPTDC/rtl/top/mptdc_{core}.sv:32-260
spadmic_{tdc_{shared}_{readout}}	Shared TDC acquisition-record readout	TOP/rtl/spadmic_{tdc_{shared}_{readout}.sv
spadmic_{position_{block}}	Async position sampling, cluster extraction, frame FIFO, packetizer, SPAD reset control	TOP/rtl/spadmic_{position_{block}.sv:11-578
spadmic_{correlated_{tx}}	Packet arbiter, event-ID tagger, post-arbiter FIFO	TOP/rtl/spadmic_{correlated_{tx}.sv:11-216
spadmic_{ddr_{tx}}	16-bit logical stream to 8-bit DDR source-synchronous PHY	TOP/rtl/spadmic_{ddr_{tx}.sv:11-45

3.1 Top-Level Port Contract

Port	Dir.	Width	Description	Source
clk_{sys}	input	1	Primary digital system clock	TOP/rtl/spadmic_{top_{v1}.sv:12
clk_{ref_{40m}}	input	1	Reference clock used by SPAD stop qualifier path	TOP/rtl/spadmic_{top_{v1}.sv:13; TOP/rtl/spadmic_{tdc_{axis_{wrapper}.sv:49-57
async_{rst_{n}}	input	1	Async active-low reset; synchronized into local reset domains	TOP/rtl/spadmic_{top_{v1}.sv:14,128-133
i2c_{scl_{i}}	input	1	I2C serial clock sampled into clk_{sys}	TOP/rtl/spadmic_{top_{v1}.sv:16; I2C/rtl/spadmic_{i2c_{slave}.sv:98-100
i2c_{sda_{i}}	input	1	I2C SDA input sampled into clk_{sys}	TOP/rtl/spadmic_{top_{v1}.sv:17; I2C/rtl/spadmic_{i2c_{slave}.sv:98-100
i2c_{sda_{oe_{o}}}	output	1	Open-drain style SDA output-enable drive	TOP/rtl/spadmic_{top_{v1}.sv:18; I2C/rtl/spadmic_{i2c_{slave}.sv:161-185
spad_{x/y/z_{event_{async_{i}}}}	input	3 x 1	Asynchronous SPAD event inputs for X/Y/Z TDC axes	TOP/rtl/spadmic_{top_{v1}.sv:20-22
cal_{x/y/z_{start_{async_{i}}}}	input	3 x 1	Calibration START async inputs	TOP/rtl/spadmic_{top_{v1}.sv:23,25,27
cal_{x/y/z_{stop_{async_{i}}}}	input	3 x 1	Calibration STOP async inputs	TOP/rtl/spadmic_{top_{v1}.sv:24,26,28
x/y/z_{lines_{i}}	input	3 x 127	Asynchronous SPAD matrix position line buses	TOP/rtl/spadmic_{top_{v1}.sv:29-31; TOP/rtl/spadmic_{pkg}.sv:19
chip_{tx_{clk_{o}}}	output	1	Forwarded source-synchronous output clock	TOP/rtl/spadmic_{top_{v1}.sv:33; TOP/rtl/spadmic_{ddr_{tx}.sv:28-31
chip_{tx_{valid_{o}}}	output	1	SDR valid for each logical 16-bit word	TOP/rtl/spadmic_{top_{v1}.sv:34; TOP/rtl/spadmic_{ddr_{tx}.sv:36-42
chip_{tx_{data_{o}}}	output	8	DDR physical TX byte lane	TOP/rtl/spadmic_{top_{v1}.sv:35; TOP/rtl/spadmic_{ddr_{tx}.sv:36-42
spad_{matrix_{rst_{o}}}	output	1	One-cycle active-high SPAD matrix reset pulse	TOP/rtl/spadmic_{top_{v1}.sv:36; TOP/rtl/spadmic_{position_{block}.sv:325-329
tdc_{stop_{armed_{o}}}	output	3	Per-axis stop qualifier armed status	TOP/rtl/spadmic_{top_{v1}.sv:38; TOP/rtl/spadmic_{tdc_{axis_{wrapper}.sv:51-57
tdc_{shared_{busy_{o}}}	output	1	Shared TDC readout busy status	TOP/rtl/spadmic_{top_{v1}.sv:39; TOP/rtl/spadmic_{tdc_{shared_{readout}.sv
position_{busy_{o}}	output	1	Position detector/packetizer busy status	TOP/rtl/spadmic_{top_{v1}.sv:40; TOP/rtl/spadmic_{position_{block}.sv:141-145

4 Configuration and Register Map

The TOP CSR address is 12 bits. Address bits [11:8] select GLOBAL, TDC_{X}, TDC_{Y}, TDC_{Z}, or POSITION. Per-axis MPTDC registers reuse the MPTDC local offsets behind unique TOP region bases, so the extracted TOP-visible

map has no duplicate addresses.

Address	Register	Access	Reset / default	Description	Source
0x000	GLOBAL_{ID}	R	See bitfield/reset table; not fully documented for all bits	constant 0x5350_{4144} (SPAD)	TOP/docs/02_{CSR}_{MAP.md:29-36; TOP/rtl/spadmic_{global_{csr.sv:134-140}
0x004	GLOBAL_{VERSION}	R	See bitfield/reset table; not fully documented for all bits	top-level version 0x0003_{0000} in active RTL	TOP/docs/02_{CSR}_{MAP.md:29-36; TOP/rtl/spadmic_{global_{csr.sv:138-140}
0x008	GLOBAL_{CTRL}	R/W	See bitfield/reset table; not fully documented for all bits	requested control image	TOP/docs/02_{CSR}_{MAP.md:38-50; TOP/rtl/spadmic_{global_{csr.sv:101-113,142-149}
0x00C	GLOBAL_{STATUS}	R	See bitfield/reset table; not fully documented for all bits	active state, datapath status, accept state	TOP/docs/02_{CSR}_{MAP.md:51-70; TOP/rtl/spadmic_{global_{csr.sv:151-167}
0x010	GLOBAL_{FAULT}	R/W1C	See bitfield/reset table; not fully documented for all bits	sticky faults	TOP/docs/02_{CSR}_{MAP.md:71-79; TOP/rtl/spadmic_{global_{csr.sv:118-121,169-174}
0x014	GLOBAL_{FAULT_{COUNT}	R	See bitfield/reset table; not fully documented for all bits	mode reject counter	TOP/docs/02_{CSR}_{MAP.md:80-85; TOP/rtl/spadmic_{global_{csr.sv:176-178}
0x400	POS_{CTRL}	R/W	See bitfield/reset table; not fully documented for all bits	position enable, packet mode, reset mode/request	TOP/docs/02_{CSR}_{MAP.md:102-123; TOP/rtl/spadmic_{position_{block.sv:317-330,513-517}
0x404	POS_{GAP_{CFG}	R/W	See bitfield/reset table; not fully documented for all bits	zero-gap threshold	TOP/docs/02_{CSR}_{MAP.md:125-130; TOP/rtl/spadmic_{position_{block.sv:332-334,519-521}
0x408	POS_{FILTER_{CFG}	R/W	See bitfield/reset table; not fully documented for all bits	minimum cluster span and settle cycles	TOP/docs/02_{CSR}_{MAP.md:131-137; TOP/rtl/spadmic_{position_{block.sv:336-339,523-526}
0x40C	POS_{RESET_{CFG}	R/W	See bitfield/reset table; not fully documented for all bits	SPAD matrix auto-reset period	TOP/docs/02_{CSR}_{MAP.md:138-148; TOP/rtl/spadmic_{position_{block.sv:341-345,528-530}
0x420	POS_{STATUS}	R	See bitfield/reset table; not fully documented for all bits	position detector and packet status	TOP/docs/02_{CSR}_{MAP.md:150-165; TOP/rtl/spadmic_{position_{block.sv:532-544}
0x424	POS_{EVENT_{COUNT}	R	See bitfield/reset table; not fully documented for all bits	accepted position-event count	TOP/docs/02_{CSR}_{MAP.md:166-170; TOP/rtl/spadmic_{position_{block.sv:546-548}
0x428	POS_{FAULT_{STATUS}	R/W1C	See bitfield/reset table; not fully documented for all bits	position sticky faults and detector state	TOP/docs/02_{CSR}_{MAP.md:172-179; TOP/rtl/spadmic_{position_{block.sv:347-352,550-554}
0x42C	POS_{DROP_{COUNT}	R	See bitfield/reset table; not fully documented for all bits	queue-full drop count	TOP/docs/02_{CSR}_{MAP.md:180-184; TOP/rtl/spadmic_{position_{block.sv:556-558}
0x430	POS_{REJECT_{COUNT}	R	See bitfield/reset table; not fully documented for all bits	glitch/empty rejection count	TOP/docs/02_{CSR}_{MAP.md:186-190; TOP/rtl/spadmic_{position_{block.sv:560-562}
0x100	TDC_{X_{CTRL}	R/W	See MPTDC reset defaults; not all read-only values reset-local	conv_{arm level and self-clearing fifo_{clr/soft_{rst pulses	MPTDC/docs/03_{CSR_{MAP.md:48-62; MPTDC/rtl/readout/mptdc_{csr_{minimal.sv:104-108,150-152; region decode TOP/docs/02_{CSR}_{MAP.md:13-23}
0x104	TDC_{X_{MODE}	R/W	See MPTDC reset defaults; not all read-only values reset-local	reserved bit, input source, output format	MPTDC/docs/03_{CSR_{MAP.md:70-81; MPTDC/rtl/readout/mptdc_{csr_{minimal.sv:110-113,154-157; region decode TOP/docs/02_{CSR}_{MAP.md:13-23}

Address	Register	Access	Reset / default	Description	Source
0x108	TDC_{X}_{MAX}_{HITS}	R/W	See MPTDC reset defaults; not all read-only values reset-local	max hits per conversion	MPTDC/docs/03_{CSR}_{MAP}.md: 83-93; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 115-117, 159-161; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x10C	TDC_{X}_{WDT}_{CTX}	R/W	See MPTDC reset defaults; not all read-only values reset-local	fast-domain context watchdog timeout	MPTDC/docs/03_{CSR}_{MAP}.md: 94-105; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 119-121, 163-165; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x110	TDC_{X}_{WDT}_{GLOBAL}	R/W	See MPTDC reset defaults; not all read-only values reset-local	system-domain global watchdog timeout	MPTDC/docs/03_{CSR}_{MAP}.md: 106-115; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 123-125, 167-169; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x120	TDC_{X}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	ready, busy, context and drain state	MPTDC/docs/03_{CSR}_{MAP}.md: 118-147; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 172-178; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x124	TDC_{X}_{HIT}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	last hit count and close flags	MPTDC/docs/03_{CSR}_{MAP}.md: 148-156; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 180-183; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x128	TDC_{X}_{FIFO}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	FIFO level/full/empty	MPTDC/docs/03_{CSR}_{MAP}.md: 157-166; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 185-189; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x12C	TDC_{X}_{WDT}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	global watchdog trip counter	MPTDC/docs/03_{CSR}_{MAP}.md: 167-174; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 191-193; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x130	TDC_{X}_{CONV}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	conversion count	MPTDC/docs/03_{CSR}_{MAP}.md: 175-182; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 195-197; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x134	TDC_{X}_{OVF}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	rejected START count	MPTDC/docs/03_{CSR}_{MAP}.md: 183-194; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 199-201; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x200	TDC_{Y}_{CTRL}	R/W	See MPTDC reset defaults; not all read-only values reset-local	conv_{arm} level and self-clearing fifo_{clr} soft_{rst} pulses	MPTDC/docs/03_{CSR}_{MAP}.md: 48-62; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 104-108, 150-152; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x204	TDC_{Y}_{MODE}	R/W	See MPTDC reset defaults; not all read-only values reset-local	reserved bit, input source, output format	MPTDC/docs/03_{CSR}_{MAP}.md: 70-81; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 110-113, 154-157; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23

Address	Register	Access	Reset / default	Description	Source
0x208	TDC_{Y}_{MAX}_{HITS}	R/W	See MPTDC reset defaults; not all read-only values reset-local	max hits per conversion	MPTDC/docs/03_{CSR}_{MAP}.md: 83-93; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 115-117, 159-161; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x20C	TDC_{Y}_{WDT}_{CTX}	R/W	See MPTDC reset defaults; not all read-only values reset-local	fast-domain context watchdog timeout	MPTDC/docs/03_{CSR}_{MAP}.md: 94-105; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 119-121, 163-165; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x210	TDC_{Y}_{WDT}_{GLOBAL}	R/W	See MPTDC reset defaults; not all read-only values reset-local	system-domain global watchdog timeout	MPTDC/docs/03_{CSR}_{MAP}.md: 106-115; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 123-125, 167-169; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x220	TDC_{Y}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	ready, busy, context and drain state	MPTDC/docs/03_{CSR}_{MAP}.md: 118-147; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 172-178; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x224	TDC_{Y}_{HIT}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	last hit count and close flags	MPTDC/docs/03_{CSR}_{MAP}.md: 148-156; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 180-183; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x228	TDC_{Y}_{FIFO}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	FIFO level/full/empty	MPTDC/docs/03_{CSR}_{MAP}.md: 157-166; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 185-189; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x22C	TDC_{Y}_{WDT}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	global watchdog trip counter	MPTDC/docs/03_{CSR}_{MAP}.md: 167-174; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 191-193; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x230	TDC_{Y}_{CONV}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	conversion count	MPTDC/docs/03_{CSR}_{MAP}.md: 175-182; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 195-197; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x234	TDC_{Y}_{OVF}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	rejected START count	MPTDC/docs/03_{CSR}_{MAP}.md: 183-194; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 199-201; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x300	TDC_{Z}_{CTRL}	R/W	See MPTDC reset defaults; not all read-only values reset-local	conv_{arm} level and self-clearing fifo_{clr} soft_{rst} pulses	MPTDC/docs/03_{CSR}_{MAP}.md: 48-62; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 104-108, 150-152; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23
0x304	TDC_{Z}_{MODE}	R/W	See MPTDC reset defaults; not all read-only values reset-local	reserved bit, input source, output format	MPTDC/docs/03_{CSR}_{MAP}.md: 70-81; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 110-113, 154-157; region decode TOP/docs/02_{CSR}_{MAP}.md: 13-23

Address	Register	Access	Reset / default	Description	Source
0x308	TDC_{Z}_{MAX}_{HITS}	R/W	See MPTDC reset defaults; not all read-only values reset-local	max hits per conversion	MPTDC/docs/03_{CSR}_{MAP}.md: 83-93; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv:115-117, 159-161; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x30C	TDC_{Z}_{WDT}_{CTX}	R/W	See MPTDC reset defaults; not all read-only values reset-local	fast-domain context watchdog timeout	MPTDC/docs/03_{CSR}_{MAP}.md: 94-105; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv:119-121, 163-165; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x310	TDC_{Z}_{WDT}_{GLOBAL}	R/W	See MPTDC reset defaults; not all read-only values reset-local	system-domain global watchdog timeout	MPTDC/docs/03_{CSR}_{MAP}.md: 106-115; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv:123-125, 167-169; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x320	TDC_{Z}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	ready, busy, context and drain state	MPTDC/docs/03_{CSR}_{MAP}.md: 118-147; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 172-178; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x324	TDC_{Z}_{HIT}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	last hit count and close flags	MPTDC/docs/03_{CSR}_{MAP}.md: 148-156; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 180-183; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x328	TDC_{Z}_{FIFO}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	FIFO level/full/empty	MPTDC/docs/03_{CSR}_{MAP}.md: 157-166; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 185-189; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x32C	TDC_{Z}_{WDT}_{STATUS}	R	See MPTDC reset defaults; not all read-only values reset-local	global watchdog trip counter	MPTDC/docs/03_{CSR}_{MAP}.md: 167-174; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 191-193; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x330	TDC_{Z}_{CONV}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	conversion count	MPTDC/docs/03_{CSR}_{MAP}.md: 175-182; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 195-197; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23
0x334	TDC_{Z}_{OVF}_{COUNT}	R	See MPTDC reset defaults; not all read-only values reset-local	rejected START count	MPTDC/docs/03_{CSR}_{MAP}.md: 183-194; MPTDC/rtl/readout/mptdc_{csr}_{minimal}.sv: 199-201; region decode TOP/docs/02_{CSR}_{MAP}.md:13-23

4.1 Selected Bitfield Detail

Register	Bits	Field	Access	Reset	Description	Source
GLOBAL_{CTRL}	[0]	req_{global}_{enable}	R/W	0	requested top-level enable	TOP/docs/02_{CSR}_{MAP}.md:42-50; TOP/rtl/spadmic_{global}_{csr}.sv:87-94,142-149
GLOBAL_{CTRL}	[3:1]	req_{axis}_{enable}	R/W	3'b111	requested enables for X/Y/Z TDC axes	TOP/docs/02_{CSR}_{MAP}.md:42-50; TOP/rtl/spadmic_{global}_{csr}.sv:87-94

Register	Bits	Field	Access	Reset	Description	Source
GLOBAL_{CTRL}	[4]	req_{position_ }enable	R/W	1	requested position block enable	TOP/docs/02_ }CSR_{MAP. md:42-50; TOP/ rtl/spadmic_ }global_{csr. sv:87-94
GLOBAL_{CTRL}	[5]	req_{shared_{tx_ }sel	R/W	SPADMIC_ {TX_{TDC	export selector	TOP/docs/02_ }CSR_{MAP. md:42-50; TOP/ rtl/spadmic_ }global_{csr. sv:87-94
GLOBAL_{CTRL}	[6]	req_{tdc_{input_ }sel	R/W	INPUT_ {SPAD	0=SPAD, 1=CAL	TOP/docs/02_ }CSR_{MAP. md:42-50; TOP/ rtl/spadmic_ }global_{csr. sv:87-94
GLOBAL_{CTRL}	[8:7]	req_{tdc_{out_ }mode	R/W	OUT_ {MODE_ {RAW_ {FEATURES	0=RAW_{FEATURES, 1=RAW_ {TIMESTAMP, 2=FULL	TOP/docs/02_ }CSR_{MAP. md:42-50; TOP/ rtl/spadmic_ }global_{csr. sv:87-94
POS_{CTRL}	[0]	local_{enable	R/W	1	local enable inside position block	TOP/docs/02_ }CSR_{MAP. md:114-123; TOP/ rtl/spadmic_ {position_{block. sv:268-276,317-330
POS_{CTRL}	[1]	pos_{mode	R/W	SPADMIC_ {POS_ {MODE_ {CLUSTER	0=cluster packet, 1=raw bitmap packet	TOP/docs/02_ }CSR_{MAP. md:114-123; TOP/ rtl/spadmic_ {position_{block. sv:317-330,475-506
POS_{CTRL}	[3:2]	spad_{reset_{mode	R/W	manual only	0>manual, 1=event-deferred, 2=peri- odic	TOP/docs/02_ }CSR_{MAP. md:114-123; TOP/ rtl/spadmic_ {position_{block. sv:358-394
POS_{CTRL}	[4]	manual_{reset_{req	W pulse/ read 0	0	write 1 emits one active-high spad_ {matrix_{rst_{o pulse	TOP/docs/02_ }CSR_{MAP. md:114-123; TOP/ rtl/spadmic_ {position_{block. sv:325-329
MPTDC_{CTRL}	[0]	conv_{arm	R/W	0	latched arm level	MPTDC/docs/03_ }CSR_{MAP. md:48-62; MPT- DC/rtl/readout/ mptdc_{csr_ }minimal.sv:87-96, 104-108
MPTDC_{CTRL}	[1]	fifo_{clr	W self- clearing	0	one-cycle synchronous FIFO clear pulse	MPTDC/docs/03_ }CSR_{MAP. md:48-62; MPT- DC/rtl/readout/ mptdc_{csr_ }minimal.sv:104- 108
MPTDC_{CTRL}	[2]	soft_{rst	W self- clearing	0	one-cycle software reset pulse	MPTDC/docs/03_ }CSR_{MAP. md:48-62; MPT- DC/rtl/top/ mptdc_{top_ }asic.sv:85-98
MPTDC_{MODE}	[1]	input_{sel	R/W	INPUT_ {SPAD	0=SPAD inputs, 1=CAL inputs	MPTDC/docs/03_ }CSR_{MAP. md:70-81; MPT- DC/rtl/readout/ mptdc_{csr_ }minimal.sv:110- 113,154-157
MPTDC_{MODE}	[3:2]	out_{mode	R/W	RAW_ {FEATURES	0=RAW_{FEATURES, 1=RAW_ {TIMESTAMP, 2=FULL	MPTDC/docs/ 03_{CSR_ }MAP.md:70- 81; MPTDC/rtl/ readout/mptdc_ {narrow16_{tx_ }v2.sv:100-135
MPTDC_{MAX_ {HITS	[3:0]	max_{hits	R/W	15	1..15 supported; 0 disabled compare	MPTDC/docs/03_ }CSR_{MAP. md:83-93; MPT- DC/rtl/ctrl/ mptdc_{meas_ }ctrl.sv:141-147

5 I2C Control-Plane Specification

Item	Confirmed value	Source	Confidence
Slave address	7'h42	TOP/rtl/spadmic_{}pkg.sv:21-25; I2C/rtl/spadmic_{}i2c_{}slave.sv:11-13	High
CSR address width	12 bits	TOP/rtl/spadmic_{}pkg.sv:21; I2C/rtl/spadmic_{}i2c_{}slave.sv:20-24	High
CSR data width	32 bits	TOP/rtl/spadmic_{}pkg.sv:22; MPTDC/rtl/pkg/mptdc_{}pkg.sv:214-215	High
Pointer format	Two bytes; high byte contributes [11:8], low byte contributes [7:0]	I2C/rtl/spadmic_{}i2c_{}slave.sv:188-220	High
Write data order	D0=[31:24], D1=[23:16], D2=[15:8], D3=[7:0]	I2C/rtl/spadmic_{}i2c_{}slave.sv:255-358	High
Read error behavior	Read response is zeroed when CSR response error is asserted	I2C/rtl/spadmic_{}i2c_{}slave.sv:365-371	High
Outstanding transactions	Single outstanding CSR transaction	I2C/rtl/spadmic_{}i2c_{}csr_{}bridge.sv:52-87	High

6 Operating Modes

Only modes found in RTL enums, CSR fields, package constants, or documentation are listed. The former MPTDC FIRST_{}HIT mode is not listed as active because the package documents its removal; max_{}hits=1 is the current fast-close mechanism.

Mode	Selection	Behavior	Output	Source
SPAD input	MPTDC MODE.input_{}sel=0 or TOP GLOBAL_{}CTRL[6]=0	uses SPAD async input path	selected by out_{}mode	MPTDC/docs/03_{}CSR_{}MAP.md:70-81; MPTDC/rtl/pkg/mptdc_{}pkg.sv:122-125
CAL input	MPTDC MODE.input_{}sel=1 or TOP GLOBAL_{}CTRL[6]=1	uses calibration START/STOP inputs	selected by out_{}mode	MPTDC/docs/03_{}CSR_{}MAP.md:223-231; TOP/rtl/spadmic_{}tdc_{}axis_{}wrapper.sv:43-47
RAW_{}FEATURES	out_{}mode=0	hit words carry nslow/nfast plus phase features	2 hit words per hit	MPTDC/rtl/pkg/mptdc_{}pkg.sv:116-120; MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:114-124,201-209
RAW_{}TIMESTAMP	out_{}mode=1	hit word W1 carries t_{}raw_{}ps[15:0]	2 hit words per hit	MPTDC/rtl/pkg/mptdc_{}pkg.sv:116-120; MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:125-131
FULL	out_{}mode=2	features plus t_{}raw_{}ps extra word	3 hit words per hit	MPTDC/rtl/pkg/mptdc_{}pkg.sv:116-120; MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:201-218
TDC-only export	tx_{}sel=TDC and position_{}enable=0	correlated TX selects TDC stream	MPTDC 16-bit packet stream patched/tagged by TOP	TOP/docs/02_{}CSR_{}MAP.md:86-97; TOP/rtl/spadmic_{}pkg.sv:173-181
Position-only export	tx_{}sel=POSITION and position_{}enable=1	correlated TX selects position stream	cluster or raw position packet	TOP/docs/02_{}CSR_{}MAP.md:86-97; TOP/rtl/spadmic_{}correlated_{}tx.sv:97-110
Both-active correlated export	tx_{}sel=TDC and position_{}enable=1	round-robin packet arbitration between TDC and position	shared 16-bit packet stream with per-source EOC ID	TOP/docs/02_{}CSR_{}MAP.md:86-97; TOP/rtl/spadmic_{}correlated_{}tx.sv:111-142,190-211

Mode	Selection	Behavior	Output	Source
Position cluster packet	POS_{}CTRL[1]=0	cluster extractor emits fixed 12-word packet	header, subheader, three axis summaries, two clusters per axis, EOC	TOP/docs/02_{}CSR_{}MAP.md:114-123; TOP/rtl/spadmic_{}position_{}block.sv:487-506
Position raw bitmap packet	POS_{}CTRL[1]=1	raw synchronized line bitmaps are emitted	1 header + 24 payload words + EOC	TOP/rtl/spadmic_{}pkg.sv:30-31; TOP/rtl/spadmic_{}position_{}block.sv:475-486
SPAD reset manual-only	POS_{}CTRL[3:2]=0	manual write pulse only	not applicable	TOP/docs/02_{}CSR_{}MAP.md:114-148; TOP/rtl/spadmic_{}position_{}block.sv:358-394
SPAD reset event-deferred	POS_{}CTRL[3:2]=1	period expiry waits for safe idle before pulse	not applicable	TOP/docs/02_{}CSR_{}MAP.md:138-148; TOP/rtl/spadmic_{}position_{}block.sv:257-263,358-376
SPAD reset periodic	POS_{}CTRL[3:2]=2	period expiry emits pulse immediately	not applicable	TOP/docs/02_{}CSR_{}MAP.md:138-148; TOP/rtl/spadmic_{}position_{}block.sv:378-388

7 Interfaces and Protocols

Interface	Signals	Protocol	Timing / assumption	Source
Top clocks/resets	clk_{}sys, clk_{}ref_{}40m, async_{}rst_{}n	clk_{}sys sampled logic; async reset synchronized in glue; ref clock qualifies stop	clk_{}sys 160 MHz from package; ref timing otherwise macro/system contract	TOP/rtl/spadmic_{}top_{}v1.sv:11-15; TOP/docs/10_{}TAPEOUT_{}READINESS.md:57-72
I2C pins	i2c_{}scl_{}i, i2c_{}sda_{}i, i2c_{}sda_{}oe_{}o	7-bit addressed pointer-based I2C sampled in clk_{}sys	2FF pin synchronization; max rate must be checked vs clk_{}sys sampling	I2C/rtl/spadmic_{}i2c_{}slave.sv:11-30,98-100; TOP/docs/10_{}TAPEOUT_{}READINESS.md:61-62
CSR	valid/write/addr/wdata/ready/rvalid/rdata/err	single request/response, writes immediate, reads wait with timeout in decoder	clk_{}sys	TOP/rtl/spadmic_{}csr_{}decoder.sv:93-129,165-213; I2C/rtl/spadmic_{}i2c_{}csr_{}bridge.sv:45-53
TDC shared acquisition export	acq_{}valid_{}o/acq_{}data_{}o/acq_{}ready_{}i	ready/valid acquisition record stream	clk_{}sys	MPTDC/rtl/top/mptdc_{}top_{}asic.sv:60-65; TOP/rtl/spadmic_{}tdc_{}axis_{}wrapper.sv:59-87
Position stream	pos_{}valid_{}o/pos_{}data_{}o/pos_{}ready_{}i	16-bit ready/valid packet stream	clk_{}sys	TOP/rtl/spadmic_{}position_{}block.sv:27-35, 471-506
Chip TX	chip_{}tx_{}clk_{}o, chip_{}tx_{}valid_{}o, chip_{}tx_{}data_{}o[7:0]	source-synchronous DDR data with SDR valid; no off-chip ready	low byte on rising clk_{}sys, high byte on falling clk_{}sys	TOP/rtl/spadmic_{}ddr_{}tx.sv:24-42

Interface	Signals	Protocol	Timing / assumption	Source
SPAD matrix line/reset	<code>x/y/z_{}lines_{}i[126:0]</code> , <code>spad_{}matrix_{}rst_{}o</code>	multibit async sampled bus with settle <code>clk_{}sys</code> sampled, matrix filter; one-cycle active-high reset pulse	contract TBD	TOP/rtl/spadmic_{}position_{}block.sv:65-73, 132-152,303-313; TOP/docs/10_{}TAPEOUT_{}READINESS.md:109-119

8 Datapath and Event Format

Stage	Operation	Output	Source
TDC input path	SPAD event or CAL pulses are gated by global/axis enable; SPAD STOP is qualified on <code>clk_{}ref_{}40m</code> .	Per-axis MPTDC start/stop async inputs	TOP/rtl/spadmic_{}tdc_{}axis_{}wrapper.sv:43-87
MPTDC measurement	Async frontend, oscillators, PD matrix, context capture, and measurement FSM produce frozen context snapshots.	Context-bank snapshot	MPTDC/rtl/top/mptdc_{}core.sv:62-247
MPTDC drain	System-clock drain FSM emits META then HIT acquisition records.	<code>mptdc_{}acq_{}rec_{}t</code> stream	MPTDC/rtl/ctrl/mptdc_{}drain_{}ctrl.sv:12-20,130-155,170-210
TDC serialization	Narrow serializer emits header, hit words, and EOC in <code>RAW_{}FEATURES</code> , <code>RAW_{}TIMESTAMP</code> , or <code>FULL</code> mode.	16-bit ready/valid TDC words	MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:100-135
Position capture	Synchronized 127-bit X/Y/Z lines are settle-filtered, scanned, queued, and packetized.	16-bit position packet stream	TOP/rtl/spadmic_{}position_{}block.sv:132-156,238-263, 397-506
Correlation/export	Correlated TX arbitrates packet-atomically and replaces EOC payload with per-source event ID.	Shared 16-bit logical stream	TOP/rtl/spadmic_{}correlated_{}tx.sv:63-76,140-142,190-211
Physical TX	DDR TX emits low byte on rising <code>clk_{}sys</code> and high byte on falling <code>clk_{}sys</code> .	Forwarded clock, valid, 8-bit DDR data	TOP/rtl/spadmic_{}ddr_{}tx.sv:24-42

8.1 Packet Field Summary

Format item	Field layout / meaning	Source
MPTDC header	[15:14]=10, [13:12]= <code>ctx_{}id</code> , [11]=phase0, [10:7]= <code>hit_{}count</code> , [6:3]=flags, [2:1]= <code>out_{}mode</code> , [0]= <code>slow_{}boundary_{}inc</code>	MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:100-112
MPTDC hit W0	[15]=0, [14:8]= <code>nslow</code> , [7:1]= <code>nfast</code> , [0]=0	MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:114-118
MPTDC features W1	[15]=0, [14:11]= <code>ns</code> , [10:7]= <code>nf</code> , [6:0]=reserved; active 8-tap values are 0..7	MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:119-124
MPTDC timestamp word	<code>t_{}raw_{}ps[15:0]</code> for <code>RAW_{}TIMESTAMP</code> and <code>FULL</code> timestamp payload	MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:125-131
MPTDC EOC	[15:14]=11, [13:0]= <code>conv_{}count</code>	MPTDC/rtl/readout/mptdc_{}narrow16_{}tx_{}v2.sv:133-135
Position cluster packet	12 words: header, subheader, X/Y/Z summaries, two cluster words per axis, EOC	TOP/rtl/spadmic_{}position_{}block.sv:487-506
Position raw packet	26 words: raw header, 8 words each for X/Y/Z line buses, EOC	TOP/rtl/spadmic_{}pkg.sv:30-31; TOP/rtl/spadmic_{}position_{}block.sv:475-486

9 Reset, Clocking, and CDC

Boundary / signal	Mechanism	Risk / signoff action	Source
<code>async_{}rst_{}n</code> -> <code>clk_{}sys</code>	<code>mptdc_{}reset_{}sync</code> STAGES=2 in TOP glue	Check reset-tree constraints and recovery/removal.	TOP/rtl/spadmic_{}top_{}v1.sv:128-133
<code>async_{}rst_{}n</code> / <code>soft_{}rst</code> -> MPTDC local reset	Soft reset holds <code>local_{}async_{}rst_{}n</code> low for multiple <code>clk_{}sys</code> cycles before synchronizer.	Software must reprogram reset-cleared CSRs after <code>soft_{}rst</code> .	MPTDC/rtl/top/mptdc_{}top_{}asic.sv:85-114

Boundary / signal	Mechanism	Risk / signoff action	Source
I2C pins -> clk_{sys}	SCL/SDA double-synchronized before START/STOP and bit decode.	Validate maximum I2C rate versus clk_{sys} sampling window.	I2C/rtl/spadmic_{i2c_{slave.sv:61-100, 125-128}}
SPAD line buses -> clk_{sys}	2FF sync plus third stability sample and settle filter.	Multibit CDC waiver and matrix settle/glitch tests required.	TOP/rtl/spadmic_{position_{block.sv:65-73, 132-152,303-313}}
MPTDC context drain -> clk_{sys}	2FF sync; static context data read after frozen data protocol.	Requires protocol proof/CDC waiver.	MPTDC/rtl/top/mptdc_{core.sv:126-128,203-216}
Oscillator generated clocks	Real macro contract not available in current RTL database.	Critical signoff blocker; generated clocks and uncertainty required.	TOP/docs/10_{TAPEOUT_{READINESS.md:121-131}}
chip_{tx_{}}* output	Source-synchronous DDR using forwarded clk_{sys}.	Receiver setup/hold and output delay constraints still required.	TOP/rtl/spadmic_{ddr_{tx.sv:24-42; TOP/docs/10_{TAPEOUT_{READINESS.md:70-72}}}}

10 Verification, Metrics, and Readiness

Verification collateral is present and documented, but this documentation refresh did not rerun simulation, lint, synthesis, or coverage. Coverage and pass/fail statements below are limited to checked-in documentation and tracked reports.

Metric	Value	Condition	Source	Conf.
TOP source-synchronous TX physical data width	8 bits	chip_{tx_{}}data_{o} width	TOP/rtl/spadmic_{pkg.sv:24}	High
TOP internal logical packet width	16 bits	mptdc_{pkg::NARROW_{}}W	MPTDC/rtl/pkg/mptdc_{pkg.sv:202-205}	High
TOP output FIFO depth	2048 words	SPADMIC_{OUTPUT_{}}FIFO_{DEPTH}	TOP/rtl/spadmic_{pkg.sv:29}	High
Position line bus width	127 lines per axis	SPADMIC_{LINE_{}}W	TOP/rtl/spadmic_{pkg.sv:19}	High
Position queue depth	16 frames	SPADMIC_{POS_{}}QUEUE_{DEPTH}	TOP/rtl/spadmic_{pkg.sv:27}	High
MPTDC phase detector geometry	8 x 8, PD_{N=64}	NE=8, PD_{N=NE*NE}	MPTDC/rtl/pkg/mptdc_{pkg.sv:29-38}	High
MPTDC FIFO depth	64 acquisition records	FIFO_{DEPTH} parameter	MPTDC/rtl/pkg/mptdc_{pkg.sv:83-85}	High
MPTDC max hits	15	MAX_{HITS} localparam	MPTDC/rtl/pkg/mptdc_{pkg.sv:79-80}	High
MPTDC system clock frequency	160 MHz / 6250 ps	package CLK_{SYS_{}}HZ/SYS_{CLK_{PS}}	MPTDC/rtl/pkg/mptdc_{pkg.sv:29-42}	High
Latest MPTDC Genus WNS	+2.6 ps	wc_{view, macro-budgeted / latest rerun}	MPTDC/lab_{snapshots/README.md:29-34}	High
Latest MPTDC total area	48523.417 um ²	Genus latest rerun snapshot	MPTDC/lab_{snapshots/README.md:29-34}	High
Latest MPTDC dynamic power	6.350 mW	Genus latest rerun snapshot; estimate	MPTDC/lab_{snapshots/README.md:29-34}	Medium
MPTDC merged IMC coverage	70.08% (11486/16389), grade 82.05%	most recent lab-server report documented in project status	MPTDC/docs/09_{PROJECT_{STATUS.md:157-161}}	Medium

10.1 Known Issues and Open Actions

Severity	Issue	Impact	Next action	Evidence
Critical	Oscillator-domain STA is not signoff-meaningful with current oscillator stub.	Cannot claim final mixed-signal timing closure.	Freeze oscillator macro contract and generated-clock constraints.	MPTDC/lab_{snapshots/README.md:9-15,95-104}
Critical	PLL, SPAD matrix, and MPTDC oscillator macro contracts are not final.	Top-level signoff and output timing budgets remain open.	Collect pinout, timing, reset, jitter, macro abstracts, and integration constraints.	TOP/docs/10_{TAPEOUT_{READINESS.md:95-131}}
Major	Latest sys-domain WNS is only +2.6 ps.	Small timing margin may degrade after place/route extraction.	Re-evaluate after Innovus and optimize FIFO/readout cone if needed.	MPTDC/lab_{snapshots/README.md:178-186}

Severity	Issue	Impact	Next action	Evidence
Major	TOP tapeout readiness requires more reset/mode-switch and negative-path closure.	Functional integration risk remains before handoff.	Run/extend TOP readiness regression, especially I2C negative paths and reset transitions.	TOP/docs/10_{}TAPEOUT_{}{}READINESS.md:14-24,156-160
Major	Merged coverage target is not closed.	Coverage closure is insufficient for final verification signoff.	Expand CSR/top/reset tests and rerun Cadence coverage.	MPTDC/docs/09_{}{}PROJECT_{}{}STATUS.md:157-161,210-215
Major	Static-data CDC and intentional async measurement structures require waiver/frozen-data review.	CDC signoff cannot rely on structural tools alone.	Prepare CDC waiver deck with protocol proof.	TOP/docs/10_{}{}TAPEOUT_{}{}READINESS.md:57-72,140-154
Minor	I2C pointer persists across STOP; software must write pointer before read or use repeated START.	Software ambiguity if undocumented.	Document host access rule in software driver/spec.	I2C/rtl/spadmic_{}i2c_{}slave.sv:114-121,139-143,169-180
Informational	Legacy TOP modules remain in-tree but are documented outside the active chip datapath.	Can confuse readers or build scripts.	Keep clearly tagged or move to legacy area when stable.	TOP/docs/10_{}{}TAPEOUT_{}{}READINESS.md:51-55

11 Traceability Matrix

Feature	Implementation	Configuration	Verification evidence	Status
I2C control plane	spadmic_{}i2c_{}slave, spadmic_{}i2c_{}csr_{}bridge, spadmic_{}csr_{}decoder	All TOP-visible CSR regions	TOP I2C unit/end-to-end tests documented	Implemented, test evidence found
Global mode sequencing	spadmic_{}global_{}csr, spadmic_{}top_{}sequencer	GLOBAL_{}CTRL, GLOBAL_{}STATUS, GLOBAL_{}FAULT	Sequencer unit and CSR stress tests documented	Implemented, test evidence found
Three TDC axes	spadmic_{}tdc_{}axis_{}wrapper x3, mptdc_{}top_{}asic x3	TDC_{}X/Y/Z regions	MPTDC unit/integration/VIP suites documented	Implemented, test evidence found
Position path	spadmic_{}position_{}block and spadmic_{}axis_{}cluster_{}scan x3	POS_{}* registers	Stress position and raw correlated TX tests documented	Implemented, test evidence found
Correlated export	spadmic_{}correlated_{}tx and spadmic_{}ddr_{}tx	GLOBAL_{}CTRL export selection	Correlated TX and DDR unit tests documented	Implemented, test evidence found
On-chip calibration LUT	Not found in active RTL	Not found	Offline calibration scripts/data only	Not found in current database
Final mixed-signal macro signoff	Placeholder contracts only	Not applicable	Open signoff actions documented	Partially implemented / blocked on macro contracts

12 Final Consistency Review

Check	Result	Basis
Top module identification	Pass	TOP filelist and readiness graph both identify spadmic_{}top_{}v1 as active top.
Register address uniqueness	Pass	Extracted TOP-visible map has no duplicate external addresses.
CSR width consistency	Pass	TOP CSR data width references MPTDC CSR_{}DATA_{}W=32.
Mode coherence	Pass	TOP and MPTDC mode enums/CSR fields agree for active modes.
Missing or blocked data	Explicitly reported	Macro contracts, oscillator timing, CDC waivers, coverage closure, and output timing remain open.
Unsupported claims	Controlled	Report avoids claiming final tapeout or mixed-signal signoff.

Area	Confidence	Reason
Top module and hierarchy	High	Filelist, active graph, and RTL instantiations agree.
Ports and interfaces	High	Taken directly from RTL declarations and assignments.
Register map	High	Cross-checked against CSR docs, package constants, and read/write RTL.
I2C protocol	High	State machine and bridge behavior are visible in RTL.
Modes	High	Enums, CSR fields, and packetizers agree for listed active modes.
Metrics	Medium	Structural values are direct RTL; synthesis/calibration values are report/document sourced.
Verification status	Medium	Collateral exists, but no regression was rerun for this refresh.

Area	Confidence	Reason
Tapeout readiness	Medium	Open issues are well documented; final macro contracts are not present.

A Appendix A: Testbench Inventory

Testbench	Area	Purpose / note	Status note	Source
tb_{spadmic_{axis_{cluster_{scan_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{axis_{cluster_{scan_{unit.sv
tb_{spadmic_{correlated_{tx_{raw_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{correlated_{tx_{raw_{unit.sv
tb_{spadmic_{correlated_{tx_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{correlated_{tx_{unit.sv
tb_{spadmic_{ddr_{tx_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{ddr_{tx_{unit.sv
tb_{spadmic_{i2c_{control_{plane_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{i2c_{control_{plane_{unit.sv
tb_{spadmic_{output_{characterization.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{output_{characterization.sv
tb_{spadmic_{ref_{stop_{qualifier_{hold_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{ref_{stop_{qualifier_{hold_{unit.sv
tb_{spadmic_{ref_{stop_{qualifier_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{ref_{stop_{qualifier_{unit.sv
tb_{spadmic_{shared_{tx_{mux_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{shared_{tx_{mux_{unit.sv
tb_{spadmic_{stress_{arbiter.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{stress_{arbiter.sv
tb_{spadmic_{stress_{cluster_{scan.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{stress_{cluster_{scan.sv
tb_{spadmic_{stress_{csr.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{stress_{csr.sv
tb_{spadmic_{stress_{position.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{stress_{position.sv
tb_{spadmic_{stress_{stop_{qualifier.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{stress_{stop_{qualifier.sv
tb_{spadmic_{tdc_{arbiter3_{unit.sv}}}	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST_{CATALOG.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST_{CATALOG.md:1-68; TOP/tb/tb_{spadmic_{tdc_{arbiter3_{unit.sv

Testbench	Area	Purpose / note	Status note	Source
tb_{spadmic}_{tdc}_{shared}_{readout}_{unit}.sv	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST}_{CATALOG}.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST}_{CATALOG}.md:1-68; TOP/tb/tb_{spadmic}_{tdc}_{shared}_{readout}_{unit}.sv
tb_{spadmic}_{top}_{sequencer}_{unit}.sv	TOP	Inferred from filename; detailed checks in TOP/docs/04_{TEST}_{CATALOG}.md	Documented active/legacy in test catalog; not rerun by this extraction	TOP/docs/04_{TEST}_{CATALOG}.md:1-68; TOP/tb/tb_{spadmic}_{top}_{sequencer}_{unit}.sv
tb_{context}_{bank}_{unit}.MPTDC.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/unit/tb_{context}_{bank}_{unit}.sv
tb_{input}_{mux}_{unit}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/unit/tb_{input}_{mux}_{unit}.sv
tb_{narrow16}_{tx}_{v2}_{unit}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/unit/tb_{narrow16}_{tx}_{v2}_{unit}.sv
tb_{reset}_{sync}_{unit}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/unit/tb_{reset}_{sync}_{unit}.sv
tb_{watchdog}_{unit}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/unit/tb_{watchdog}_{unit}.sv
tb_{async}_{kernel}_{semantics}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{async}_{kernel}_{semantics}.sv
tb_{backpressure}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{backpressure}.sv
tb_{cal}_{inject}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{cal}_{inject}.sv
tb_{campaign}_{collect}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{campaign}_{collect}.sv
tb_{deadtime}_{measure}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{deadtime}_{measure}.sv
tb_{firsthit}_{mode}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{firsthit}_{mode}.sv
tb_{local}_{export}_{boundary}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{local}_{export}_{boundary}.sv
tb_{multi}_{conv}_{stress}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{multi}_{conv}_{stress}.sv
tb_{overflow}_{count}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{overflow}_{count}.sv

Testbench	Area	Purpose / note	Status note	Source
tb_{single}_{conv}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{single}_{conv}.sv
tb_{start}_{wdt}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{start}_{wdt}.sv
tb_{watchdog}_{recovery}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/int/tb_{watchdog}_{recovery}.sv
mptdc_{vip}_{tb}.sv	MPTDC	Inferred from filename and MPTDC verification docs	Documented maintained suite; not rerun by this extraction	MPTDC/docs/09_{PROJECT}_{STATUS}.md:157-170; MPTDC/tb/tests/mptdc_{vip}_{tb}.sv

B Appendix B: Dataset Package

Item	Value
Machine-readable JSON	\{\texttt{rtl}\}_{_}\{extracted\}_{_}\{dataset.json\}
Machine-readable CSV	\{\texttt{rtl}\}_{_}\{extracted\}_{_}\{dataset.csv\}
PDF source	\{\texttt{rtl}\}_{_}\{current\}_{_}\{state\}_{_}\{spec.tex\}
PDF output	\{\texttt{rtl}\}_{_}\{current\}_{_}\{state\}_{_}\{spec.pdf\}
Extraction notes	\{\texttt{extraction}\}_{_}\{notes.md\}

C Appendix C: Acronyms

Acronym	Definition
ASIC	Application-Specific Integrated Circuit
CSR	Control and Status Register
CDC	Clock Domain Crossing
DDR	Double Data Rate
EOC	End of Conversion
FSM	Finite State Machine
I2C	Inter-Integrated Circuit serial bus
MPTDC	Multi-phase Time-to-Digital Converter
PD	Phase Detector
PLL	Phase-Locked Loop
SPAD	Single-Photon Avalanche Diode
STA	Static Timing Analysis
TDC	Time-to-Digital Converter
WNS	Worst Negative Slack